CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2016 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Differentiate between Database System and File System.**

Differences: DBMS offers centralized control, eliminated data inconsistency, program-data independence, and built-in transaction recovery, which file systems lack.

Q2 (3M) **Explain Foreign Key with ON DELETE CASCADE.**

Explanation: A foreign key links a child record to a parent record. With `ON DELETE CASCADE`, deleting a parent row automatically cascades and deletes all corresponding child rows, maintaining referential integrity.

Q3 (3M) **What is a Weak Entity Set? How is it identified?**

Definition: An entity set lacking sufficient attributes to form a primary key. Identified using a *discriminator (partial key)* combined with the primary key of its identifying strong entity set.

Q4 (3M) **Differentiate between Natural Join and Theta Join.**

Differences: Theta Join (R \bowtie_\theta S) combines rows based on any arbitrary condition. Natural Join (R \bowtie S) automatically equates attributes having identical names and eliminates duplicate columns.

Q5 (3M) **Explain COMMIT and ROLLBACK commands in SQL.**

TCL Commands:

- `COMMIT` : Permanently saves all transaction modifications to disk.
- `ROLLBACK` : Reverts all uncommitted changes back to the start of the transaction or savepoint.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain BCNF (Boyce-Codd Normal Form) with example.**

Explanation: A relation is in BCNF if for every non-trivial functional dependency X \to Y, X is a Super Key. Eliminates anomalies arising from overlapping candidate keys.

Q7 (7.5M) **Compare Dense Index and Sparse Index in database indexing.**

Comparison:

- **Dense Index:** Contains an index record for *every single search-key value* in the data file. Faster search, larger index size.
- **Sparse Index:** Contains index records for only some search-key values (typically one per disk block). Consumes less space, but requires scanning within blocks.

Q8 (7.5M) **Explain Deadlock Detection and Prevention methods in DBMS.**

Methods: Deadlock prevention uses timestamp ordering (Wait-Die and Wound-Wait schemes). Deadlock detection periodically constructs a Wait-For Graph (WFG) and aborts a cycle-causing transaction.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Draw a detailed ER Diagram for a Banking Enterprise.**

Entities: `CUSTOMER`, `ACCOUNT`, `BRANCH`, `LOAN`.
Relationships: Customer deposits in Account, Customer borrows Loan, Branch maintains Accounts and Loans.

Q10 (15M) **Explain Relational Algebra operators: Selection, Projection, Cartesian Product, Join, and Division.**

Relational Algebra: Complete explanation of fundamental algebraic query operators with mathematical notations and practical examples.

Q11 (15M) **Explain Nested Subqueries and Correlated Subqueries in SQL with examples.**

Subqueries:

- **Nested Subquery:** Inner query executes once, and its result is used by outer query (e.g., `WHERE Salary > (SELECT AVG(Salary) FROM Emp)`).
- **Correlated Subquery:** Inner query references columns of the outer query and re-executes for each row evaluated by outer query.

Q12 (15M) **Explain ACID Properties and Serializability in transaction management.**

ACID & Serializability: In-depth coverage of Atomicity, Consistency, Isolation, Durability, conflict serializability, and conflict equivalence of schedules.

Q13 (15M) **Explain ISAM and B+ Tree File Organizations with comparative analysis.**

File Organization: Detailed architectural comparison between static ISAM hierarchy and dynamic self-balancing B+ Trees.
